

IMPLEMENTATION_PLAN.md

SUMMARY

Goal: Build TeamSync incrementally. Each phase should be independently testable, deployable, small enough for AI coding agents, and avoid large context windows. Never generate the entire application at once. Each phase becomes the foundation for the next. Estimated Timeline: 8–12 Weeks.

PHASE 0 - PROJECT FOUNDATION

Goal: Setup the entire project structure before writing business logic. Deliverables include Frontend (React + Vite + TypeScript, Material UI Setup, Framer Motion Setup, React Query Setup, Axios Instance, React Router Setup, Folder Structure, Theme Configuration) and Backend (Spring Boot Project, Maven Dependencies, MySQL Connection, Environment Variables, Global Exception Handler, DTO Structure, Mapper Structure, Response Wrapper). Folder Structure Frontend: src/, components/, pages/, layouts/, hooks/, services/, types/, utils/, context/, theme/. Backend: controller/, service/, repository/, entity/, dto/, mapper/, security/, config/, exception/. Acceptance Criteria: Application starts successfully. Frontend and backend communicate.

PHASE 1 - AUTHENTICATION SYSTEM

Goal: Build complete authentication. Database: users. Features: Register, Login, Logout, JWT, Role Storage, Password Encryption, Spring Security. Frontend Screens: Login Page, Register Page, Protected Route. Backend APIs: POST /auth/register, POST /auth/login. Deliverables: JWT Authentication, BCrypt Password Hashing, Role Storage, Auth Context, Token Persistence. Acceptance Criteria: User can register. User can login. Protected routes work.

PHASE 2 - APPLICATION LAYOUT

Goal: Build reusable application shell. Features: Sidebar, Top Navbar, Profile Menu, Theme Toggle, Notification Icon, Global Search UI. Layouts: Auth Layout, Dashboard Layout. Deliverables: Responsive Navigation, Dark Mode, Page Routing. Acceptance Criteria: Every future screen uses same layout.

PHASE 3 - USER PROFILE MODULE

Goal: User profile management. Database: users. Additional Fields: avatar_url, department, bio. Features: View Profile, Edit Profile, Upload Avatar, Cloudinary Integration. APIs: GET /users/profile, PUT /users/profile, POST /users/avatar. Acceptance Criteria: User can manage profile. Cloudinary uploads work.

PHASE 4 - PROJECT MODULE

Goal: Project creation and management. Database: projects. Features: Create Project, Edit Project, Delete Project, Project Details, Project Listing, Project Status. APIs: POST /projects, GET /projects, PUT /projects/{id}, DELETE /projects/{id}. Frontend: Project List, Project Form, Project Details. Acceptance Criteria: Projects fully functional.

PHASE 5 - PROJECT MEMBERS MODULE

Goal: Team collaboration begins. Database: project_members. Features: Add Members, Remove Members, Assign Lead, View Team. APIs: POST /projects/{id}/members, DELETE /projects/{id}/members/{id}, GET /projects/{id}/members. Acceptance Criteria: Projects can have teams.

PHASE 6 - TASK MODULE CORE

Goal: Core task management. Database: tasks. Features: Create Task, Update Task, Delete Task, Assign Task, Status Update, Priority Levels. APIs: POST /tasks, PUT /tasks, DELETE /tasks. Frontend: Task List, Task Details, Task Creation Modal. Acceptance Criteria: Tasks work end-to-end.

PHASE 7 - KANBAN BOARD

Goal: Visual task management. Libraries: DND Kit. Features: Drag & Drop, Status Changes, Column Updates, Realtime UI Updates. Acceptance Criteria: Task movement updates database.

PHASE 8 - COMMENTS MODULE

Goal: Task collaboration. Database: comments. Features: Add Comment, Edit Comment, Delete Comment, Task Discussions. APIs: POST /comments, GET /comments/task/{id}. Acceptance Criteria: Task conversations work.

PHASE 9 - ACTIVITY FEED

Goal: Track all important actions. Database: activity_logs. Track: Task Created, Task Assigned, Task Completed, Project Created, Member Added. Frontend: Activity Timeline. Acceptance Criteria: Users see project history.

PHASE 10 - FILE ATTACHMENTS

Goal: Attach files to tasks. Database: attachments. Cloudinary. Features: Upload File, Delete File, Preview File. Supported: PDF, DOCX, Images. Acceptance Criteria: Files available from task details.

PHASE 11 - NOTIFICATION SYSTEM

Goal: User awareness. Database: notifications. Triggers: Task Assigned, Task Completed, Comment Added, Member Added. Frontend: Notification Drawer, Unread Count. Acceptance Criteria: Notifications generated automatically.

PHASE 12 - SEARCH SYSTEM

Goal: Global search. Search Across: Projects, Tasks, Members, Comments. Frontend: Command Palette, Ctrl + K. Acceptance Criteria: Users find anything quickly.

PHASE 13 - ANALYTICS DASHBOARD

Goal: Project insights. Widgets: Project Health, Tasks Completed, Overdue Tasks, Member Productivity. Charts: Recharts. Acceptance Criteria: Dashboard displays meaningful metrics.

PHASE 14 - ROLE PERMISSIONS HARDENING

Goal: Security validation. Admin: Full Access. Lead: Project Access. Member: Task Access. Acceptance Criteria: Role restrictions enforced.

PHASE 15 - MOBILE RESPONSIVENESS

Goal: Production usability. Features: Responsive Layout, Responsive Tables, Responsive Drawers, Mobile Navigation. Acceptance Criteria: Works on mobile devices.

PHASE 16 - UI ENHANCEMENT

Goal: Premium SaaS experience. Features: Framer Motion, Microinteractions, Page Transitions, Loading States, Skeleton Screens, Empty States, Success Animations. Acceptance Criteria: Application feels polished.

PHASE 17 - ADVANCED SAAS EXPERIENCE

Goal: Differentiate from normal CRUD apps. Features: Command Center, Quick Actions, Spotlight Search, Project Health Visualizations, Workspace Improvements. Acceptance Criteria: Feels like a modern SaaS product.

PHASE 18 - 3D EXPERIENCE LAYER

Goal: Add visual wow-factor. Libraries: React Three Fiber, Drei. Features: Landing Hero, 3D Project Network, Team Relationship Graph, Analytics Visualization. Important: Do NOT add 3D inside productivity workflows. Acceptance Criteria: Enhanced visuals without reducing usability.

PHASE 19 - TESTING

Frontend: Component Tests. Backend: Service Tests, Repository Tests, Security Tests. Acceptance Criteria: Critical flows tested.

PHASE 20 - DEPLOYMENT

Frontend: Vercel. Backend: Render. Database: Railway MySQL. Monitoring: Sentry. Acceptance Criteria: Production deployment complete.

PHASE 21 - PORTFOLIO PREPARATION

Deliverables: README, Architecture Diagram, ER Diagram, API Documentation, Screenshots, Demo Video, Deployment Links. Acceptance Criteria: Project ready for recruiters.